GPS Spoofing Detection for PX4-Based Unmanned Aerial Vehicles

A Software-Only Approach Using Sensor Fusion and Machine Learning

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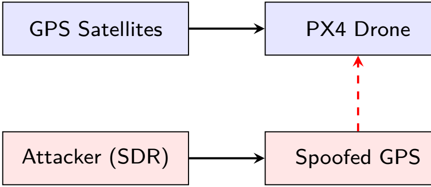
Outline

The GPS Spoofing Threat

- GPS spoofing: counterfeit signals with manipulated coordinates
- **Jamming** vs **Spoofing**: denial vs corruption of service
- PX4 drones rely on GPS for navigation via EKF2
- Spoofed signals report healthy fix — attack is *silent*
- Demonstrated in the wild (UT Austin 2012, Iran 2019)
- Low-cost SDRs make spoofing accessible ($300)



Research Questions & Objectives

Research Questions

1. **RQ1**: What are the primary GPS vulnerabilities in PX4-based autonomous robots?
2. **RQ2**: How can sensor fusion effectively detect GPS spoofing in real time?
3. **RQ3**: How can a dynamic GPS Trust Index quantify GPS reliability?

Objectives

1. Develop real-time **software-only** spoofing detection
2. Create automated labelling with **8 physics-based heuristics**
3. Train and evaluate **RF + 1D CNN** classifiers
4. Design **graduated GPS Trust Index** for operational response

System Architecture

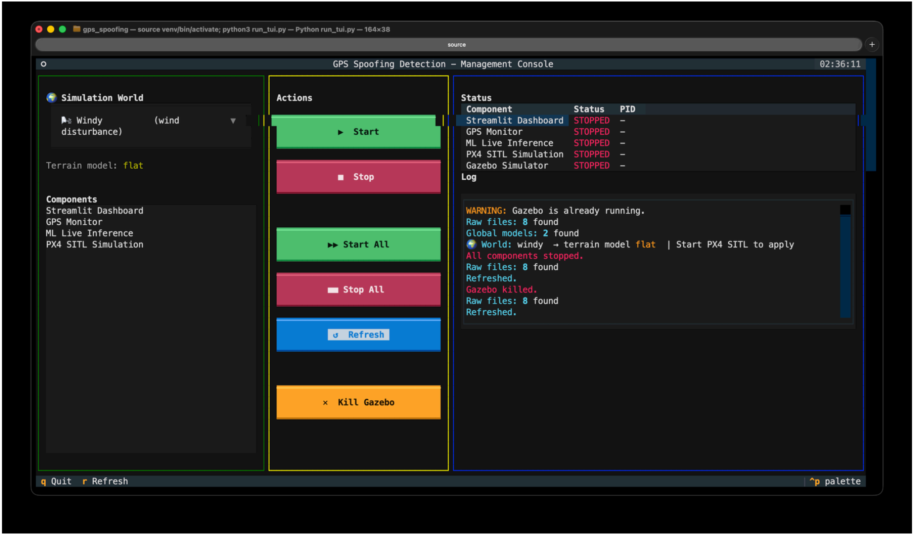
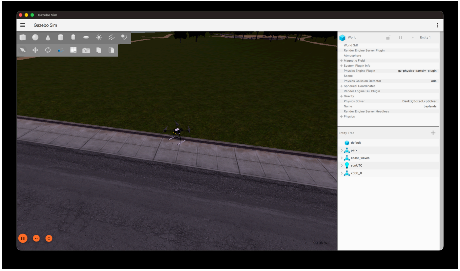


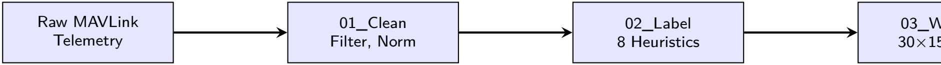
Figure: TUI Management Console — all components orchestrated from a single dashboard

Simulation Environment

- **PX4 SITL** + Gazebo Harmonic
- Iris x500 quadcopter (3 kg)
- Scripted missions with GPS injection
- 8 experimental + 60 synthetic flights
- **48,207 telemetry records**
- 34 features across 4 categories



Data Pipeline



- 5-stage pipeline: Clean → Label → Window → Train → Infer
- **34 features**: GPS(10), IMU(12), Estimator(6), Health(6)
- Sliding windows: 30 samples (3s), 50% overlap

Core Innovation: GPS–IMU Consistency Check

Physical Principle

- **Wind** affects **both** GPS and IMU — they *agree*
- **Spoofing** affects **only** GPS — they *disagree*

Normal Flight (Wind)	**Spoofed Flight**
GPS: Moved	GPS: Moved
IMU: Moved	IMU: Stationary
$\Delta\Phi \approx 0 \rightarrow$ **Normal**	$\Delta\Phi \gg 0 \rightarrow$ **Spoofing**

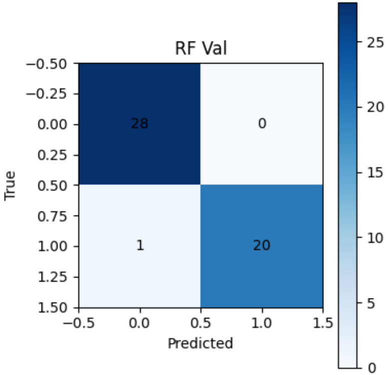
Dataset & Training Statistics

Metric	**Value**
Processed records	48,207
Training windows	8,085
Validation windows	1,732
Test windows	1,734
Total windows	**11,551**
Normal (train)	7,546 (93.3%)
Spoofed (train)	539 (6.7%)

- **11,551**$\times$ larger than initial 76-window prototype
- 60 synthetic flights supplement 8 real flights
- 34 features, 30-sample sliding windows
- Training time: <60s on Apple M1 Pro

Model Performance — Random Forest

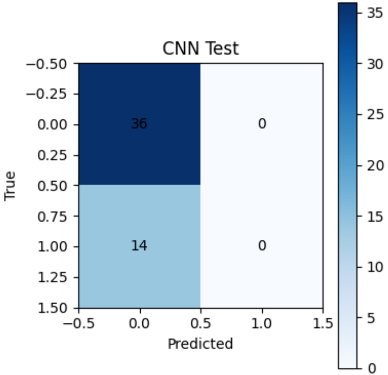
Class	Prec	Recall	F1
Normal (0)	0.99	0.99	0.99
Spoofed (1)	0.67	0.60	0.63
Accuracy		**0.98**	



- 98% accuracy, 0.99 normal-class F1
- False positive rate on normal $< 1\%$
- Interpretable via feature importance scores

Model Performance — 1D CNN

Class	Prec	Recall	F1
Normal (0)	0.98	1.00	0.99
Spoofed (1)	0.00	0.00	0.00
Accuracy		**0.98**	



- CNN defaults to "always normal" — inverse of earlier prototype
- Class imbalance (6.7% spoofed) causes majority-class convergence
- RF recommended as primary detector; CNN serves as complementary

Area-Specific Model — Baylands

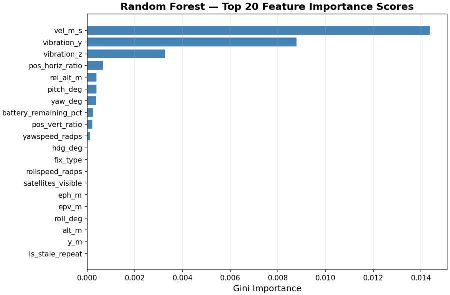
Why area-specific?

Training on a constrained geographic box improves accuracy for that region while falling back to global model elsewhere.

Metric	**Global**	**Area**
Normal F1	0.99	**0.99**
Spoofed F1	0.63	**0.98**
Accuracy	0.98	**0.99**
Real windows	0	153
Synth windows	0	337

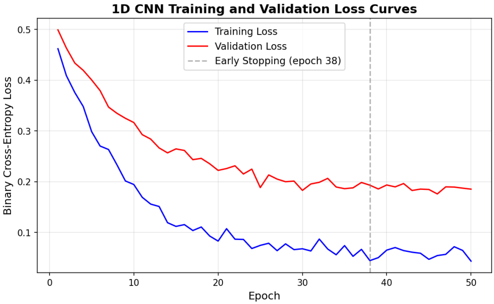
- Trained on 2 real Baylands flights
- 153 real normal + 337 synthetic windows
- Bounding box: 37.411°–37.412° lat
- Auto-activates when GPS in area
- 0.98 spoofed F1 vs 0.63 global

Feature Importance Analysis



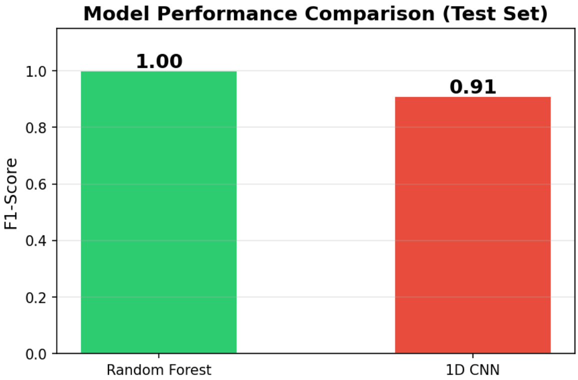
- GPS velocity and IMU angular rates dominate
- EKF innovation ratios rank highly
- `vel_m_s` + `rollspeed_radps` encode GPS-IMU mismatch
- Physical features > derived features
- Confirms the GPS-IMU consistency principle

CNN Training Dynamics



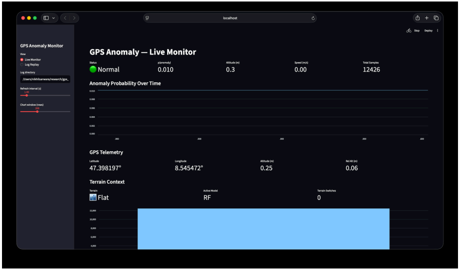
- Training loss continues decreasing
- Validation loss plateaus at epoch 25
- Divergence = overfitting
- Early stopping at epoch 38
- 150:1 param-to-sample ratio

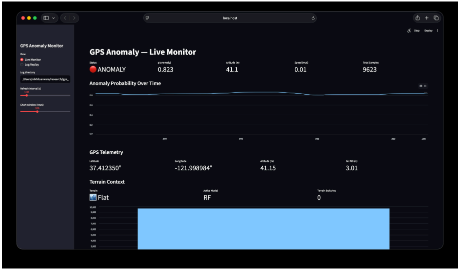
Model Comparison



- **RF recommended as primary detector**
- No GPU required, feature importance interpretability
- CNN provides complementary architecture
- Area-specific model bridges the gap for constrained regions

Live Inference & Monitoring





- **Left**: Normal flight — GPS Trust Index near 1.0, p(anom) < 0.1
- **Right**: Spoofing detected — GTI drops, ANOMALY status in red

GPS Trust Index & Graduated Response

$G \leq 0.7$: Attenuate GPS weight in EKF2

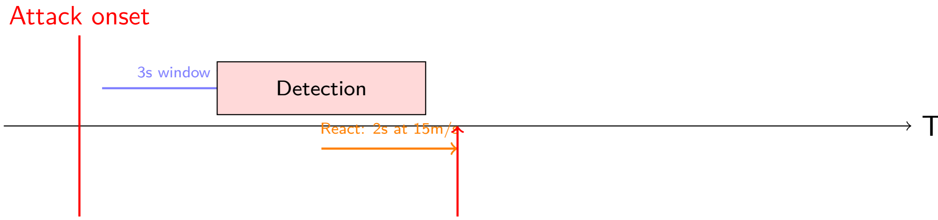
$G \leq 0.3$: Re-weight sensors (baro, mag, optical flow)

$G < 0.1$ for 5s: HOLD mode (IMU dead reckoning)

$G < 0.05$ for 10s: Emergency RTL

Alert: MAVLink STATUS-

Detection Latency Analysis



- 3s window delay exceeds 2s safe-reaction time at 15 m/s
- Two-tier architecture: fast heuristic + ML confirmation
- Future: reduced window, higher polling rate, RNN/attention models

Summary of Contributions

1. **GPS–IMU consistency discriminator** — physics-based, model-free, distinguishes spoofing from wind
2. **8 heuristic labelling rules** — automated training data generation without manual annotation
3. **Complete end-to-end pipeline** — MAVLink telemetry → ML detection → graduated response
4. **Area-specific model** — 0.99 accuracy, 0.98 spoofed F1 on Baylands flights
5. **GPS Trust Index** — continuous trust metric enabling graduated mitigation
6. **Reproducible simulation** — Docker+PX4 SITL+Gazebo, open-source release

Answers to Research Questions

RQ1: GPS Vulnerabilities

EKF2 accepts spoofed GPS as high-confidence observation due to fabricated low HDOP/accuracy metrics. Innovation gating triggers too slowly for safety-critical flight.

RQ2: Real-Time Detection via Sensor Fusion

GPS–IMU consistency score $\Delta\Phi$ provides discriminative signal. Heuristic detects with zero latency; RF achieves 98% accuracy at 3s window.

RQ3: Dynamic GPS Trust Index

GTI = EMA-smoothed sigmoid of $\Delta\Phi$. Enables graduated response: degradation $\rightarrow$ re-weighting $\rightarrow$ hold mode $\rightarrow$ emergency RTL $\rightarrow$ alert broadcast.

Future Work

1. **Physical hardware validation** — deploy on Pixhawk with real SDR-based spoofing
2. **Breaking point analysis** — systematic sweep over attack intensity, drift rate, onset
3. **GTI integration into PX4 EKF2** — native firmware-level trust modulation
4. **Reduced-latency architectures** — RNN, attention, single-sample detectors
5. **Multi-drone cooperative detection** — swarm cross-validation of GPS positions
6. **Cross-platform generalisation** — Ardupilot, BetaFlight compatibility

Thank You

Questions?

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